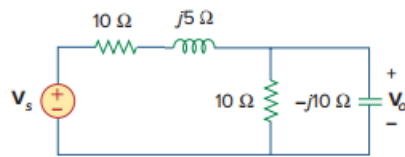


Problem

If $\mathbf{V}_o = 8\angle 30^\circ \text{ V}$ in the circuit of Fig. 9.59, find $\mathbf{I}_s$.

Figure 9.59:



Step-by-step solution

Step 1 of 3

Refer to Figure 9.59 in the textbook.

I found typo error in the question. The unknown factor in the circuit is $\mathbf{V}_s$.

Write the expression for $\mathbf{V}_o$.

$$\mathbf{V}_o = \frac{[(10) \parallel (-j10)]}{10 + j5 + [(10) \parallel (-j10)]} \mathbf{V}_s$$

Calculate the value of $(10) \parallel (-j10)$.

$$\begin{aligned} [(10) \parallel (-j10)] &= \frac{(10)(-j10)}{(10) + (-j10)} \\ &= 5 - j5 \end{aligned}$$

Step 2 of 3

Calculate the value of $\mathbf{V}_s$.

$$\mathbf{V}_o = \frac{[(10) \parallel (-j10)]}{10 + j5 + [(10) \parallel (-j10)]} \mathbf{V}_s$$

$$8\angle 30^\circ = \left(\frac{5 - j5}{10 + j5 + 5 - j5} \right) \mathbf{V}_s$$

$$8\angle 30^\circ = \left(\frac{5 - j5}{15} \right) \mathbf{V}_s$$

$$\mathbf{V}_s = \left(\frac{15}{5 - j5} \right) (8\angle 30^\circ)$$

$$\mathbf{V}_s = 16.97\angle 75^\circ \text{ V}$$

Thus, the value of $\mathbf{V}_s$ is $\boxed{16.97\angle 75^\circ \text{ V}}$.

Step 3 of 3

If the unknown factor is current through the voltage source, the current through the voltage source is $\mathbf{I}_s$.

Calculate the value of $\mathbf{I}_s$.

$$\begin{aligned}\mathbf{I}_s &= \frac{\mathbf{V}_s - \mathbf{V}_o}{10 + j5} \\ &= \frac{(16.97\angle 75^\circ) - (8\angle 30^\circ)}{10 + j5} \\ &= 1.294\angle 75^\circ \text{ A}\end{aligned}$$

Thus, the value of the current through the voltage source is $\boxed{1.294\angle 75^\circ \text{ A}}$.